

Time-Off Microservice — Requirements in Plain Terms

ExampleHR Engineering

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What Are We Building?

A backend service that lets employees request time off through **ExampleHR**, while keeping those requests in sync with an external **HR system** (called the HCM — think Workday or SAP). The HCM is the “source of truth” for how many leave days each employee actually has.

The Core Problem (In Plain English)

Imagine Alice has **10 vacation days**. She opens ExampleHR and requests 2 days off. Simple so far. But here’s what makes it hard:

1. **Two systems, one truth.** ExampleHR is the interface, but the HCM holds the real balance. If ExampleHR and the HCM disagree, bad things happen — Alice could end up with more leave than she’s entitled to, or be incorrectly told she has no leave left.
 2. **The HCM changes balances on its own.** On Alice’s work anniversary, the HCM might automatically add 5 bonus days. ExampleHR won’t know about this unless it’s told.
 3. **Two requests can arrive at the same time.** Alice and her colleague Bob both submit requests simultaneously. Together they might exceed the available balance. The system must ensure only one goes through.
 4. **The HCM is not 100% reliable.** Sometimes the HCM won’t return an error even when a request exceeds the balance. ExampleHR can’t blindly trust HCM’s error responses — it needs its own safety checks.
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What the System Must Do

For Employees

- **See an accurate balance** — available days = what the HCM says minus any requests they’ve already submitted that are waiting for manager approval.
- **Get instant feedback** — when they submit a request, they immediately know if it was accepted or rejected (insufficient balance, invalid dates, etc.).
- **Cancel requests** — before or after manager approval.

For Managers

- **Approve or reject requests** — knowing the balance shown is valid and up to date.
- **Approve with confidence** — the system does a live check with the HCM at approval time to make sure the balance hasn't changed.

For the System

- **Stay in sync with the HCM** — via two mechanisms:
 - *Real-time*: the HCM can push a single balance update (e.g., after an anniversary bonus).
 - *Batch*: the HCM can push a full snapshot of all employee balances periodically.
- **Never over-allocate leave** — even if two employees (or the same employee twice) submit requests at exactly the same moment.
- **Be defensive** — even if the HCM silently accepts an invalid request, ExampleHR catches it first.

Key Business Rules

Rule	Plain Explanation
Balances are per employee per location	Alice in London and Alice in New York have separate leave pools
Available = HCM balance - pending days	Pending means “submitted, not yet approved/rejected”
Pending reservation is immediate	The moment you submit a request, those days are reserved locally
HCM is notified only on approval	Draft/pending requests don't go to HCM; only approved ones do
Approved cancellation credits back	Cancelling an approved request tells the HCM to refund the days
Discrepancies are flagged, not auto-fixed	If HCM shows fewer days than our pending requests, a human must review

Request Lifecycle

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Employee submits > PENDING > Manager approves > APPROVED > Period ends > COMPLETED
|
|-- Manager rejects > REJECTED
|
\-- Employee cancels > CANCELLED

                                APPROVED > Employee cancels > CANCELLED
```

What the System Does NOT Do

- It does not enforce leave policies (e.g., “you can’t take more than 5 days consecutively”) — that’s the HCM’s job.
 - It does not handle weekends or public holidays in date calculations — the number of days is provided by the caller.
 - It does not send email or push notifications.
 - It does not handle multiple tenants (single company assumed).
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Who Talks to Whom

Employee / Manager

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ExampleHR API -----> HCM (Workday/SAP)
|                               |
| <<-- "How many days does Alice have?" |
| --> "Approve 2 days for Alice"         |
|                               |
| <<-- "Alice just got a 5-day bonus"    <<---
| <<-- "Here's everyone's balance"      <<---
```

Summary

In one sentence: **ExampleHR is the face, the HCM is the truth** — this service makes sure they always agree, and protects employees from ever accidentally going over their balance, no matter how fast they click.